Beyond the Score: Early Evaluations of GPT-6 Astra as a Robot Critic

Simple AI

Can a general-purpose vision–language model provide useful feedback on robot behavior? We report exploratory offline evaluations of GPT-6 Astra at low reasoning effort on trajectory assessment and visual subtask diagnosis. On the evaluated RBM data, zero-shot Astra obtains a reported OOD progress VOC of 93.6, trajectory-ranking Kendall τ-a of 0.8166, and direct pairwise preference accuracy of 92.4%. Context examples are associated with a small increase in progress VOC but a decrease in ranking, accompanied by more tied scores. On 57 RoboFail questions, the external five-example condition has higher classification accuracy than zero-shot (85.96% versus 64.91%), but lower failure recall (82.35% versus 94.12%). Approximately 89.3% of the increase in the benchmark's composite ROUGE-L score comes from correctly recognizing successful subtasks, not from its failure-explanation component. These observations motivate reporting progress, discrimination, error asymmetry, and explanation quality separately. They do not establish a causal benefit of examples, online failure localization, or improved robot learning: model-backend consistency is unconfirmed for the RoboFail comparison, and repeated-run and closed-loop evaluations remain outstanding.

Date: September 23, 2026
Status: Preliminary technical report

1 Introduction

Robot critics are intended to assess behavior, rather than directly generate actions. Useful assessment can take several forms: estimating progress from an observed prefix, assigning a score that orders trajectories, choosing between two executions, or diagnosing whether an intermediate subtask succeeded. A strong result on one of these interfaces does not establish competence on the others. In particular, correlation with elapsed time need not imply accurate completion estimates, and a higher classification score need not imply that fewer failures are missed.

This report examines these distinctions using an existing evaluation record for GPT-6 Astra. Our emphasis is on interpreting the reported measurements rather than establishing a leaderboard claim. We retain three main observations: context examples do not uniformly improve trajectory metrics; successful-subtask recognition and failure recall can move in opposite directions; and the benchmark's explanation score contains a substantial classification component. All experiments discussed below are offline. No policy is trained or controlled by the critic in this report.

Evidence boundary. This preliminary draft consolidates the study record available on September 23, 2026. Numerical results are retained from that record; confusion-matrix arithmetic is checked directly, and plots are reconstructed from its reported aggregates. We do not infer missing confidence intervals, repeat-run variance, or undocumented protocol details. Literature-only scores are presented separately as context, not pooled into our metric-wise rankings, because matched sample identities, inputs, and scoring conditions have not been established here.

2 Evaluation Scope and Protocol

Table 1 Distinct evaluation interfaces. Counts describe the evaluated record, not independent robot trials. The preference set is reported separately; its split composition is not specified in the source record.

Interface	Model input and assessment	Evaluation support
Progress	Task text and observed prefix; correlation with a time reference	105 successful trajectories
Independent ranking	Task text and eight frames from one trajectory; score ordering	567 trajectories; 41 source–task groups
Direct preference	Two trajectories in the same request; choose the better one	1,192 fixed pairs
Subtask diagnosis	Original keyframe montage and a subtask question; label and explanation	57 questions; 15 trajectory groups

2.1 Trajectory assessment

The progress and ranking study uses six sources in RBM-EVAL-OOD. Progress predictions are evaluated against the time reference using the reported VOC metric. VOC is presented on the source record's $\times 100$ scale; it is a correlation measure, not a percentage of correctly judged stages. Progress examples use five evaluation positions, with only the current and preceding frames visible at each prediction. The reported mean squared error (MSE) also uses the time-based reference, not independently annotated semantic completion percentages.

For ranking, each trajectory receives a score independently. The reference order is success above suboptimal above failure, and the reported Kendall τ-a penalizes predicted ties relative to a correct strict order. Direct pairwise preference is a different interface: both trajectories are available simultaneously. Its accuracy cannot be interpreted as the accuracy of ordering independently computed scores.

We distinguish two context-example conditions instead of calling both "one-shot." *Single-trajectory ICL* for progress provides one same- or similar-task training trajectory with six stage scores. *Three-trajectory reference ICL* for ranking provides randomly selected high-, middle-, and low-score training references without requiring a task match. The middle reference score is not the test set's suboptimal label. These conditions differ in example content and count and are not an equal-context comparison.

2.2 Visual subtask diagnosis

RoboFail, as distributed with AHA [11], provides the 57 image–question examples used here: 23 successful and 34 failed subtasks, grouped into 15 operation sequences by trajectory identifier. Zero-shot uses the original montage and question. The external five-example condition prepends a fixed set of five image–question–answer examples, reported not to overlap the test trajectories. This does not establish environment, embodiment, near-duplicate, or pretraining independence.

The study follows the AHA evaluation definitions [12]. Binary Success is label accuracy, not robot execution success. The reported ROUGE-L is a composite: correct success answers receive 1, correct failure answers receive stemmed ROUGE-L F1 against the reference explanation, and label disagreements receive 0. Scoring uses the original English answers, not translated quotations.

Comparison conditions. The model labels below retain the study's request names and reasoning settings. They do not imply equal compute, latency, or token budgets. For RoboFail, returned model identifiers differ between the two conditions despite the same requested model name and reasoning effort. Backend snapshot consistency is therefore unconfirmed. We report an observed condition difference, not an isolated causal effect of ICL. Confidence intervals and repeated-call variation were not supplied with these aggregate results; all differences below are descriptive.

3 RBM Evaluation: Performance and Diagnostics

This section evaluates progress estimation, independent trajectory scoring, and direct pairwise preference on the RBM evaluation study described in section 2. We first summarize benchmark performance, then examine why a high aggregate score does not by itself certify completion understanding.

3.1 Benchmark comparison and metric-wise standing

Table 2 RBM evaluation results only. VOC and ranking use RBM-EVAL-OOD; external preference uses its different-quality track, while this study's RBM pair split composition remains incompletely documented. [S]: this study; [P]: Robometer paper Tables XIII, XVI, XVII [1]; [W]: benchmark website detailed OOD tables [3]; [R]: RynnValue Table 2 [2]. Bold identifies Astra zero-shot, not cross-protocol statistical superiority. VOC is scaled by 100; missing results are not imputed.

Reported configuration	OOD VOC ↑	OOD τ-a ↑	Preference ↑
GPT-6 Astra low (zero-shot) [S]	**93.6**	**0.8166**	**92.4%**
GPT-6 Astra low (ICL) [S]	94.2	0.7953	—
GPT-5.6-sol low [S]	93.5	0.750	90.7%
DpskV4.1-flash max [S]	89.7	0.530	82.2%
RynnValue-8B [R]	—	0.675	—
RynnValue-4B [R]	—	0.670	—
Robometer / RBM-1M (paper v1) [P]	94.8	0.655	82.5%
Robometer / RBM-1M (web) [W]	94.0	0.643	82.5%
Robometer / RoboReward data [P]	92.7	0.552	—
RoboReward-4B [P]	87.9	0.502	—
RoboReward-8B [P]	88.2	0.465	—
RoboDopamine-8B (web) [W]	79.6	0.453	—
TOPReward / Qwen3-VL-8B [W]	39.7	0.128	—
GVL / GPT-5-mini [P]	26.2	0.199	—
VLAC-8B [P]	37.5	0.049	—
VLAC-2B [P]	—	0.077	—
ReWiND / RBM-1M [P]	50.8	0.014	—
RL-VLM-F / GPT-5-mini [P]	—	—	55.5%

GPT-6 Astra's strongest evidence is comparative judgment. Within the three tested zero-shot general-purpose configurations, Astra ranks first on all three reported metrics (table 3). Its lead over the next-best configuration, Sol, is 0.0666 in Kendall τ-a and 1.7 percentage points in preference accuracy; the VOC lead is only 0.1 point. Thus the clearest numerical separation is in independent trajectory ranking, rather than progress correlation. These are descriptive point-estimate ranks without uncertainty estimates, not significance claims or an equal-compute comparison.

Table 3 Metric-wise standing within this study's three zero-shot general-purpose configurations. ICL is displayed separately in table 2, not treated as an additional model or selectively combined with zero-shot results.

Metric	Astra (rank 1)	Sol (rank 2)	Dpsk (rank 3)
OOD VOC ↑	**93.6**	93.5	89.7
OOD Kendall τ-a ↑	**0.8166**	0.75	0.53
Direct preference ↑	**92.4%**	90.7%	82.2%

Specialist context, not a unified leaderboard. Robometer's published tables report OOD VOC 94.8, Kendall 0.655, and different-quality preference accuracy 82.5% [1]. The latter uses 500 comparisons per OOD dataset, unlike our 1,192 fixed pairs with incompletely documented split composition. RynnValue-8B reports Kendall 0.675 [2]; its paper describes a 976-trajectory OOD suite, while this study scores 567 trajectories. Exact sampling and aggregation correspondence has not been established. Astra's ranking and preference numbers are numerically higher than these available literature references, while its VOC (93.6 zero-shot; 94.2 with ICL) remains below Robometer's 94.8. We therefore report promising comparative capability, not a protocol-matched specialist victory or SOTA. Unverified RynnValue VOC and preference values have been removed. ID VOC is not ranked because this study reports no general-purpose-model ID result.

Broader published comparisons. Table 2 and figures 1 to 3 include only results explicitly linked to the RBM evaluation suite. GVL, VLAC-2B/8B, RoboReward-4B/8B, ReWiND, and RL-VLM-F are baselines evaluated by Robometer's authors; their method papers are not claimed to have originally evaluated RBM [1, 4–8]. RynnValue-4B/8B report RBM-EVAL-OOD in their own paper; TOPReward and RoboDopamine-8B have RBM-EVAL-OOD results on the benchmark authors' website [2, 3, 9, 10]. Method citations identify approaches; source tags identify the numerical evaluation records. Names such as "RoboReward data" or "RBM-1M" specify training data, not a different test benchmark. No ManiRewardBench, RoboRewardBench, or real-robot policy success rate is imported into this comparison merely because a metric has a similar name.

The current website reports Robometer OOD VOC 94.0 and Kendall 0.643, versus 94.8 and 0.655 in paper v1. Both records are shown, not merged or selected for advantage. We use the website's detailed OOD tables, whose entries are more precise than its rounded overview; some overview entries are inconsistent with those detailed tables. This is a documented literature snapshot, not an exhaustive leaderboard. Unverified RynnValue VOC/preference entries are not plotted. Robometer trained on RoboReward data and on RBM-1M are separate configurations, as is ReWiND retrained on RBM-1M. Cross-record differences remain descriptive, without matched-input or statistical superiority claims.

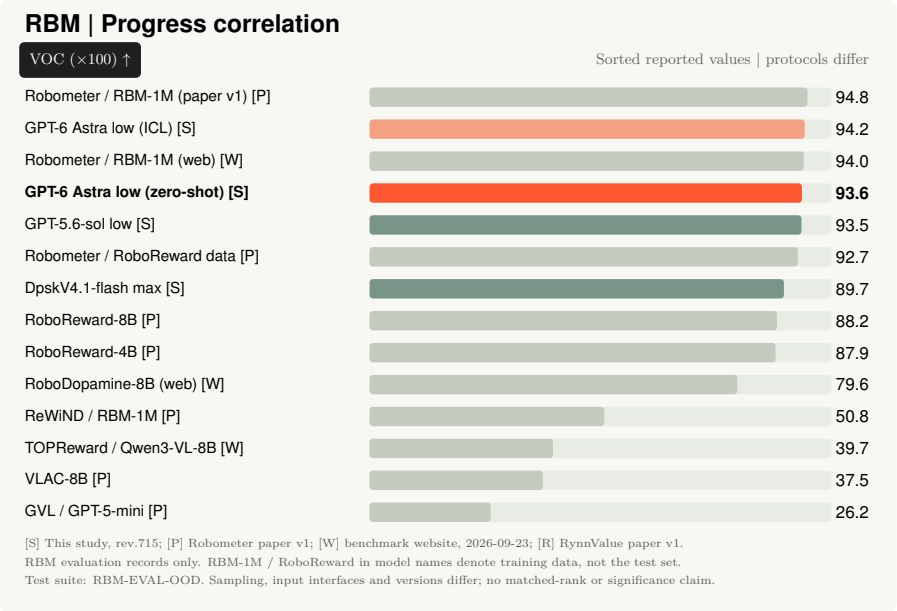


Figure 1 Progress correlation on RBM-EVAL-OOD. Bars are sorted by reported value, not protocol-matched rank. Orange highlights Astra; green denotes other study models; gray denotes external records. Missing metrics are omitted; all available values appear in table 2. Sources and versions are retained.

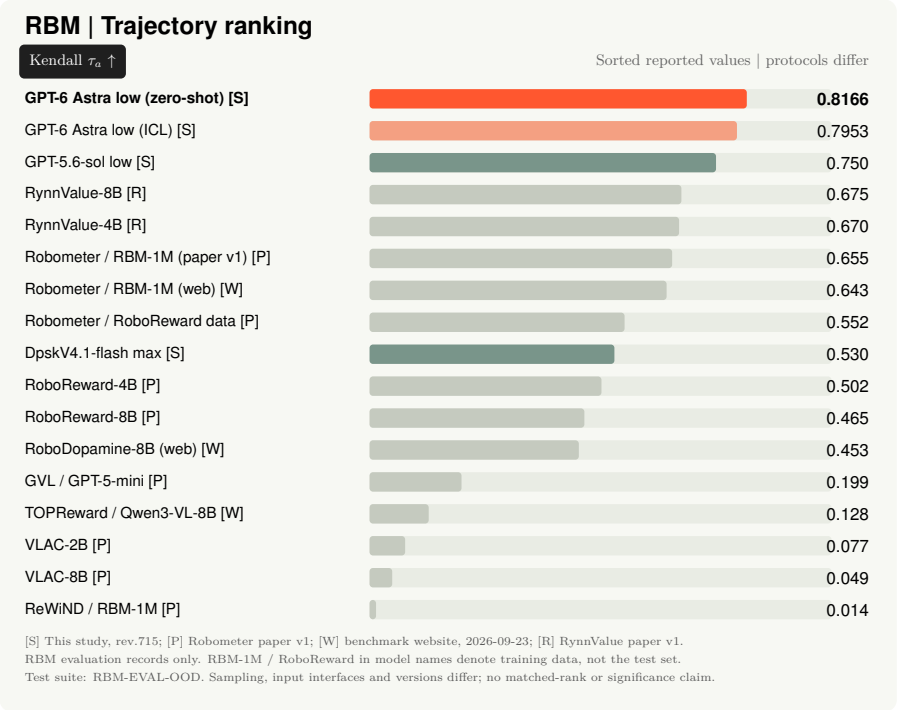


Figure 2 Trajectory ranking on RBM-EVAL-OOD. Separate reporting versions remain separate rows. Astra's numerical lead is descriptive: different trajectory samples and input interfaces prevent a matched-leaderboard or significance claim.

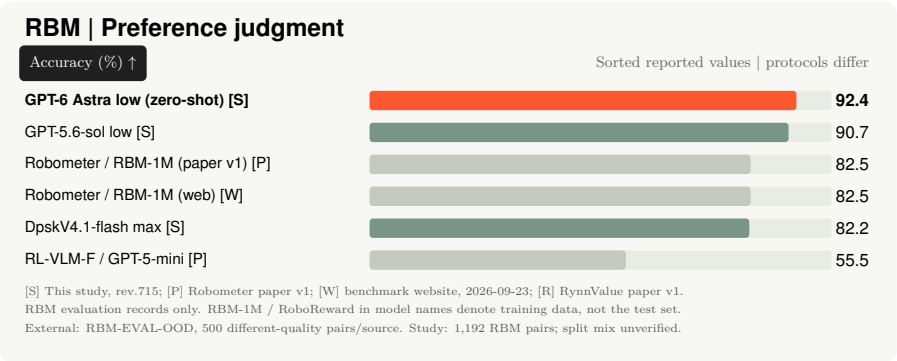


Figure 3 Direct preference judgment on RBM. External results use 500 different-quality pairs per OOD source; this study uses 1,192 fixed RBM pairs with incompletely documented split composition. These are not different-task preference results, independent-score Kendall values, or policy success rates.

3.2 Progress correlation and endpoint accuracy can disagree

With single-trajectory ICL, aggregate VOC increases from 93.6 to 94.2, and time-reference MSE decreases from 0.0402 to 0.0390. Of 105 trajectories, VOC improves on 58, declines on 35, and is unchanged on 12. This change is heterogeneous: in the reported banana-transfer task, 18 of 19 trajectories have lower VOC and one is unchanged. Moreover, 11 of the 58 trajectories with higher VOC have worse MSE.

Table 4 A reported bottle-opening example, selected post hoc. The public terminal label is success. A higher time correlation coexists with a lower terminal score and larger time-reference MSE. This is an illustrative counterexample, not an estimate of the frequency of such errors.

Condition	Terminal score	VOC (unscaled) ↑	MSE ↓
Zero-shot	1.00	0.9051	0.0407
Single-trajectory ICL	0.35	0.9536	0.1312

The bottle-opening case in table 4 makes the distinction concrete. ICL has a higher VOC while its endpoint estimate is farther from the public success label; MSE is approximately 3.22× larger. We do not independently adjudicate the visual completion state or relabel this example. The observation shows why time correlation alone cannot validate completion assessment. Likewise, an operation that remains in preparation before a late state change need not make uniform semantic progress over time.

3.3 Fewer inversions do not guarantee better ranking

Three-trajectory reference ICL changes τ-a from 0.8166 to 0.7953. Of 41 source–task groups, eight improve, ten decline, and 23 are unchanged. Under the study's score-consistent pair weighting, correct strict orders decrease from 86.44% to 83.21%, ties increase from 8.78% to 13.11%, and inversions decrease from 4.78% to 3.68% (figure 4). The arithmetic is consistent with the reported score: $0.8644 - 0.0478 = 0.8166$, versus $0.8321 - 0.0368 = 0.7953$. This decomposition concerns weighted outcomes, not unweighted counts of independent pairs.



Figure 4 Ranking decomposition reconstructed from reported aggregate weights. Reference ICL has fewer inversions but more ties and fewer correct strict orders. No error bars are shown because per-task predictions and resampling estimates are not included in this draft.

Selected disagreements also suggest that visible task completion and public quality grades may differ. They do not establish that most errors are annotation ambiguities. Without independent blind review, label ambiguity, insufficient visual evidence, and model error remain competing explanations. Public labels are retained throughout; model-generated rationales are not treated as evidence of ground-truth correctness.

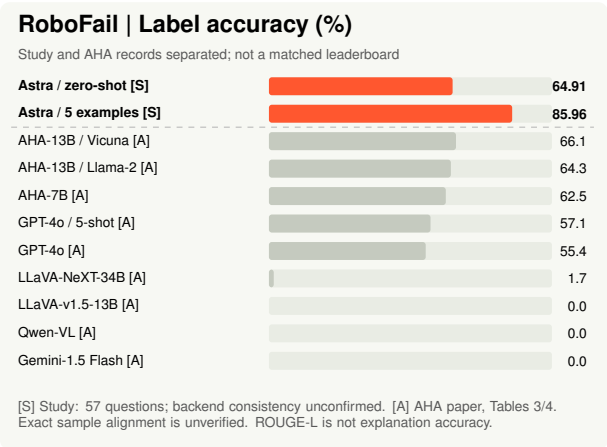
4 Subtask Diagnosis: Accuracy Conceals an Alarm Trade-off

4.1 Published RoboFail evaluations: separate adaptations, not one leaderboard

RoboFail names a source dataset, but the papers below do not all evaluate the same questions. We retain only explicitly reported RoboFail results: AHA's adapted benchmark [13], Guardian's 153 execution and 30 planning examples [14], and the 100-example Cosmos-curated split used by PRIMO [15]. Our study contains 57 questions. The latter two adaptations are therefore reported separately, without placing Astra in their rankings. No ManiSkill-Fail, UR5-Fail, RoboVQA, or PRIMO progress-benchmark score is imported into these tables.

Table 5 This study [S] and published AHA-adapted RoboFail records [A]. AHA values come from Tables 3/4 of the authors' paper [13]; fractional Binary Success entries are displayed as percentages. Exact question identity/count alignment with our 57-question record is unverified. "Astra" denotes the requested configuration, not confirmed identical returned backends across conditions. Published zeros are preserved, not interpreted as intrinsic inability.

Configuration / source	Label accuracy ↑	Reported ROUGE-L ↑
Astra / zero-shot [S]	64.91%	0.185
Astra / 5 examples [S]	85.96%	0.499
LLaVA-v1.5-13B [A]	0.00%	0.000
LLaVA-NeXT-34B [A]	1.70%	0.018
Qwen-VL [A]	0.00%	0.000
Gemini-1.5 Flash [A]	0.00%	0.000
GPT-4o [A]	55.40%	0.114
GPT-4o / 5-shot [A]	57.10%	0.236
AHA-7B [A]	62.50%	0.204
AHA-13B / Llama-2 [A]	64.30%	0.280
AHA-13B / Vicuna [A]	66.10%	0.290



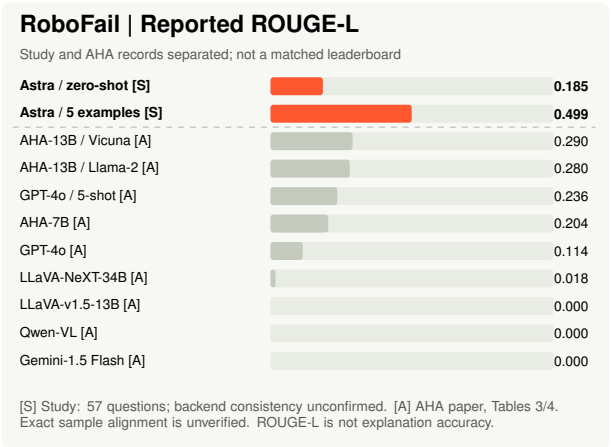


Figure 5 Label accuracy and reported ROUGE-L are distinct metrics. Orange study records and gray AHA literature records are separated, not pooled into a matched ranking. This study's composite ROUGE-L is decomposed in equation (1); the literature aggregate is transcribed, not independently re-scored. Neither bars nor numerical gaps establish explanation correctness or statistical superiority.

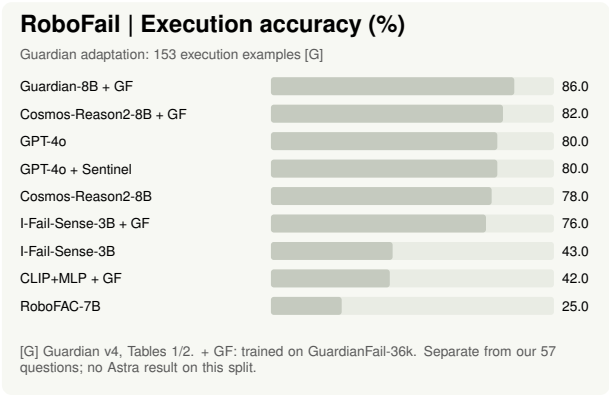
Table 6 Other verified RoboFail adaptations, reported separately. Left: Guardian v4 Tables 1/2 [14]; right: PRIMO Table 3 / Appendix A.3 [15]. All values are accuracy percentages. +GF denotes training on GuardianFail-36k. Guardian's inherited AHA score is omitted here because it is not a rerun on its 153-example execution split. No Astra score is available on either adaptation.

Guardian: 153 execution / 30 planning

Configuration	Exec	Plan
GPT-4o	80.0	67.0
GPT-4o + Sentinel	80.0	63.0
RoboFAC-7B	25.0	5.0
I-Fail-Sense-3B	43.0	67.0
Cosmos-Reason2-8B	78.0	53.0
CLIP+MLP + GF	42.0	43.0
I-Fail-Sense-3B + GF	76.0	52.0
Cosmos-Reason2-8B + GF	82.0	70.0
Guardian-8B + GF	86.0	70.0

Cosmos / PRIMO: 100 examples

Configuration	Accuracy
Gemini 2.0 Flash	67.0
GPT-4o	63.0
OpenAI o1	61.0
Claude-haiku-4.5	59.0
Qwen2.5-VL-7B	57.6
Nemotron-H-56B	64.0
Cosmos-Reason1-7B	60.0
Cosmos-Reason1-56B	66.2
PRIMO (SFT)	51.0
PRIMO (RL)	63.0
PRIMO R1	67.0



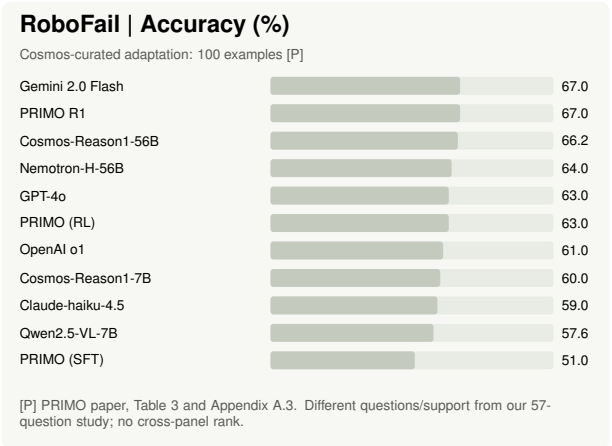


Figure 6 Two separate RoboFail adaptations. Ordering is within each published record only; the panels do not share evaluation support. Guardian's planning accuracy remains in table 6, not mixed with execution accuracy. Error bars are absent because per-example predictions and compatible uncertainty estimates have not been obtained.

The same model name can have substantially different scores across these records (e.g., GPT-4o), reinforcing the need to bind sample identities, inputs, prompts, parsers, and aggregation before claiming a shared leaderboard. AHA-13B's Vicuna variant is kept distinct from its Llama-2 variant. We do not infer missing failure recall or explanation metrics for Guardian or PRIMO. Our 85.96% five-example accuracy must not be used to claim superiority over their 86.0% or 67.0% results on different supports.

4.2 Success recognition improves while failure recall declines

We define failure as the positive class. A false alarm is a reference-success sample predicted as failure; a miss is a reference-failure sample predicted as success. This is an offline classification convention, not an online alarm with a measured timestamp.

Table 7 RoboFail results on the same 57 questions. All percentages are computed from the displayed counts. The composite ROUGE-L is rounded to three decimals as in the main study table. Backend snapshot consistency is unconfirmed; differences are observational.

Statistic	Zero-shot	External five-example condition
Correct success judgments	5/23	21/23
Correct failure judgments	32/34	28/34
Overall label accuracy ↑	37/57 (64.91%)	49/57 (85.96%)
Failure recall ↑	94.12%	82.35%
Miss rate ↓	2/34 (5.88%)	6/34 (17.65%)
False-alarm rate ↓	18/23 (78.26%)	2/23 (8.70%)
Composite ROUGE-L ↑	0.185	0.499

The five-example condition has 12 more correct answers overall: 16 more correct success judgments, offset by four fewer correct failure judgments. Thus, its higher overall accuracy does not mean that it detects more failures. Whether this trade-off is useful depends on the cost of unnecessary intervention relative to missed failures; neither deployment cost nor closed-loop performance is measured here.

4.3 Decomposing the composite explanation score

Let $y_i \in \{S, F\}$ denote the reference label, $\hat{y}_i$ the prediction, and r_i the stemmed ROUGE-L F1 for a correctly classified failure. The benchmark score reported in the study is

$$R = \frac{1}{N} \sum_{i=1}^{N} \left[\mathbf{1}(y_i = S, \hat{y}_i = S) + \mathbf{1}(y_i = F, \hat{y}_i = F) r_i \right], \qquad N = 57. \tag{1}$$

The two components are reported as 0.0877 and 0.0969 for zero-shot, versus 0.3684 and 0.1305 for the five-example condition. The success term alone increases by $16/57 \approx 0.2807$, accounting for approximately 89.3% of the reported total increase of 0.3144. The remaining approximately 10.7% comes from the failure-explanation term. Component values are rounded; arithmetic with them can differ in the last displayed decimal from totals computed before rounding.

A score of 0.499 therefore does not mean that 49.9% of failure explanations are correct. The explanation component depends both on which failures are detected and on lexical overlap. It cannot distinguish a mechanism-correct paraphrase from a lexically similar but incorrect diagnosis. A post-hoc knob-grasping example illustrates this limitation: zero-shot predicts failure, agreeing with the reference label, but describes an incorrect grasp location rather than the reference's failure of the gripper to close. The five-example condition predicts success on that same failed subtask. The example does not establish either rationale as a verified physical cause.

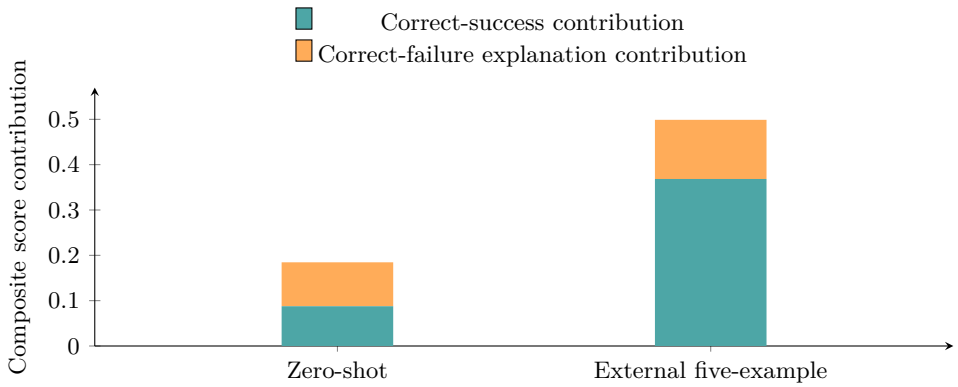


Figure 7 Arithmetic composition of the benchmark's ROUGE-L score. Both components use all 57 questions as the denominator. Most of the observed increase is in the automatically credited success term; this is a score decomposition, not a causal attribution or a semantic explanation-accuracy estimate.

5 Discussion, Limitations, and Next Validation Steps

What the evidence supports. The evaluated configurations exhibit useful offline discrimination, but their strengths depend on the interface and metric. The progress counterexample and the RoboFail decomposition both show that a favorable aggregate score can coexist with an unfavorable error relevant to critic use. Reporting only VOC, label accuracy, or composite ROUGE-L would hide these distinctions. The results motivate joint reporting of endpoint judgments, strict ranking and ties, false alarms and misses, and independently assessed explanations.

What it does not support. These experiments do not demonstrate reward-model safety, reliable contact-state reasoning, causal failure diagnosis, online localization or recovery detection, or improved reinforcement-learning sample efficiency. They also do not establish state-of-the-art performance under a matched protocol. Specialist-model literature values are not rerun baselines, and more detailed model responses are not automatically more faithful explanations.

Statistical and experimental limitations. The 57 RoboFail questions share only 15 trajectory groups, so they are not 57 independent executions. Ranking pairs can share trajectories and tasks as well. Repeated calls and sampling uncertainty are not quantified in the current record. Small differences, including the 0.6-point progress VOC change, should not be described as statistically established improvements. Future paired uncertainty estimates should preserve trajectory dependence for RoboFail and source–task dependence for RBM while retaining the intended aggregation weights. Such intervals would still not resolve the RoboFail backend confound or uncertainty from example selection. Claims about label ambiguity require independent blind annotation, not post-hoc selection of persuasive cases.

Reproducibility status. The record identifies evaluation counts, reference-example constructions, and aggregate outcomes, but does not provide all details needed for an independently reproducible release. Before publication, the accompanying materials should bind the six source identities and sample manifests; the preference split composition; actual input-frame identities and resolution; complete prompts and example identities; requested and returned model identifiers and run dates; output budgets and sampling settings; parsing, refusal, retry and missing-output rules; and scorer versions with exact aggregation and tie conventions. Per-sample predictions are needed for an independent score audit and cluster-aware uncertainty estimates. We do not invent these missing fields or treat their absence as proof that the underlying experiments were invalid.

Priorities for further validation. First, repeat the paired context comparison under a confirmed common model snapshot, with fixed inputs and example-selection rules. Second, add independent semantic assessments of completion and failure explanations, including ambiguous and low-observability cases. Third,

evaluate whether critic feedback helps policy learning using a fixed initial policy, algorithm, interaction budget, and success measure independent of the critic. The third step is future work: no RL outcome is reported here. It should measure both learning curves and feedback cost, and explicitly check reward exploitation rather than assume that stronger offline scores translate to policy gains.

6 Conclusion

These early evaluations suggest that GPT-6 Astra can provide informative offline assessments of robot trajectories and subtasks, while revealing important gaps between benchmark scores and the behaviors a critic must exhibit in practice. Context examples are not associated with uniform gains: ranking can lose resolution, and higher subtask accuracy can accompany lower failure recall. The central finding is consequently methodological rather than a deployment claim: robot-critic evaluation should expose which decisions improved, which errors increased, and which parts of a composite score changed before attributing broader competence to a model.

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